PSY 5790: Applied Univariate and Bivariate/Bivariable Statistics

Between and Within-Subject t-Tests Demos

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```
# install.packages("datarium")
# install.packages("rstatix")
# install.packages("ggpubr")

library(jmv)
library(psych)
library(tidyverse)
library(infer)
library(datarium)
library(rstatix)
library(ggpubr)
```

One-Sample `t`-test

One numerical variable

Research Question 1: Is the average weight of mice different from 25g? What about less than 25g?

```
# Load and inspect demo data set(s)

# Load `mice` data from `datarium` package
data(mice, package="datarium")
head(mice)
```

```
## # A tibble: 6 × 2
##   name  weight
##   <chr>  <dbl>
## 1 M_1     18.9
## 2 M_2     19.5
## 3 M_3     23.1
## 4 M_4     20.1
## 5 M_5     20.3
## 6 M_6     23.4
```

Hypothesis Statement: 2-tailed (Non-directional)

H_0: Null Hypothesis — Average weight of sampled mice is not different from 25 g,
$\mu_{sample} = \mu_{population} = 25 \text{ grams}$

H_a: Alternate Hypothesis — Average weight of sampled mice is different from 25 g,
$\mu_{sample} \neq \mu_{population} \neq 25 \text{ grams}$

Hypothesis Statement: 1-tailed (Directional)

H_0: Null Hypothesis — Average weight of sampled mice is not less than 25 g,
$\mu_{sample} \geq \mu_{population} \geq 25 \text{ grams}$

H_a: Alternate Hypothesis — Average weight of sampled mice is less than 25 g,
$\mu_{sample} < \mu_{population} < 25 \text{ grams}$

Research Question 2: Is the average hours of work different from 40 hours per week? What about greater than 40 hours per week?

```
set.seed(13)
# Load `gss` data from `infer` package
data(gss, package="infer")
gss <- gss %>%
  select(hours) %>%
  rep_sample_n(size = 100, replace = TRUE)
head(gss)
```

```
## # A tibble: 6 × 2
## # Groups:   replicate [1]
##   replicate hours
##       <int> <dbl>
## 1         1    40
## 2         1    64
## 3         1    70
## 4         1    55
## 5         1    50
## 6         1    48
```

Hypothesis Statement: 2-tailed (Non-directional)

H_0: Null Hypothesis — Average hours of work per week is not different from 40,
$\mu_{sample} = \mu_{population} = 40 \text{ hours per week}$

H_a: Alternate Hypothesis — Average hours of work per week is different from 40,
$\mu_{sample} \neq \mu_{population} \neq 40 \text{ hours per week}$

Hypothesis Statement: 1-tailed (Directional)

H_0: Null Hypothesis — Average hours of work per week is not greater than from 40,
$\mu_{sample} \leq \mu_{population} \leq 40 \text{ hours per week}$

H_a: Alternate Hypothesis — Average hours of work per week is different from 40,
$\mu_{sample} > \mu_{population} > 40 \text{ hours per week}$

Descriptive Statistics (Do not forget to visualize!!)

```
#Descriptive statistics for `weight` variable
describe(mice$weight)
```

```
##    vars  n  mean  sd median trimmed  mad  min  max range skew kurtosis  se
## X1    1 10 20.14 1.9   19.8   20.06 1.48 17.5 23.4   5.9 0.52    -1.08 0.6
```

```
#Descriptive statistics for `hours` variable
describe(gss$hours)
```

```
##    vars   n  mean    sd median trimmed   mad min max range skew kurtosis   se
## X1    1 100 45.39 16.14   40.5   44.94 11.12   4  89    85 0.31     1.28 1.61
```

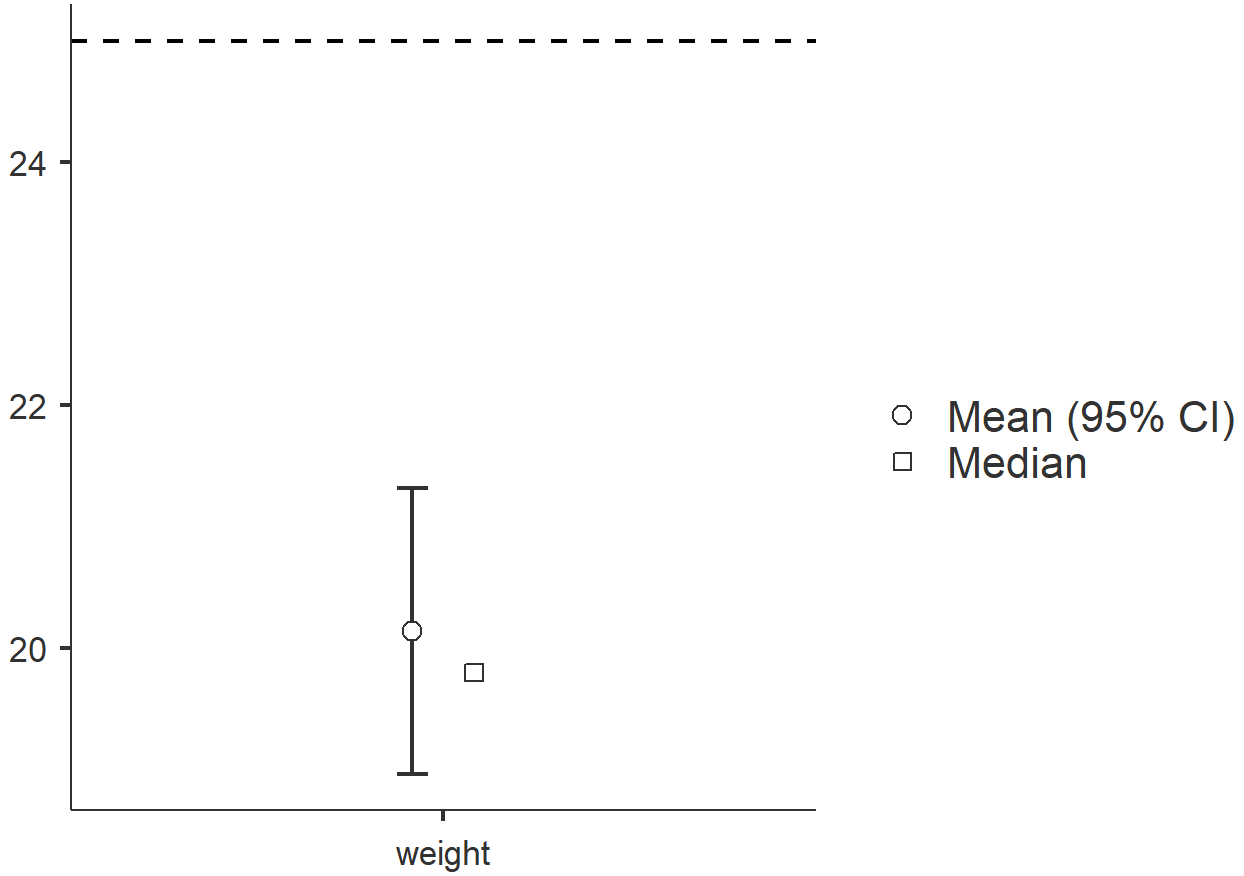
`jmv` One-Sample `t`-test `ttestOneS()`

Research Question 1

```
mice %>%
ttestOneS(vars=weight,
          testValue = 25,
          hypothesis = 'dt',
          norm = TRUE,
          ci = TRUE,
          effectSize = TRUE,
          ciES = TRUE,
          desc = TRUE,
          plots = TRUE)
```

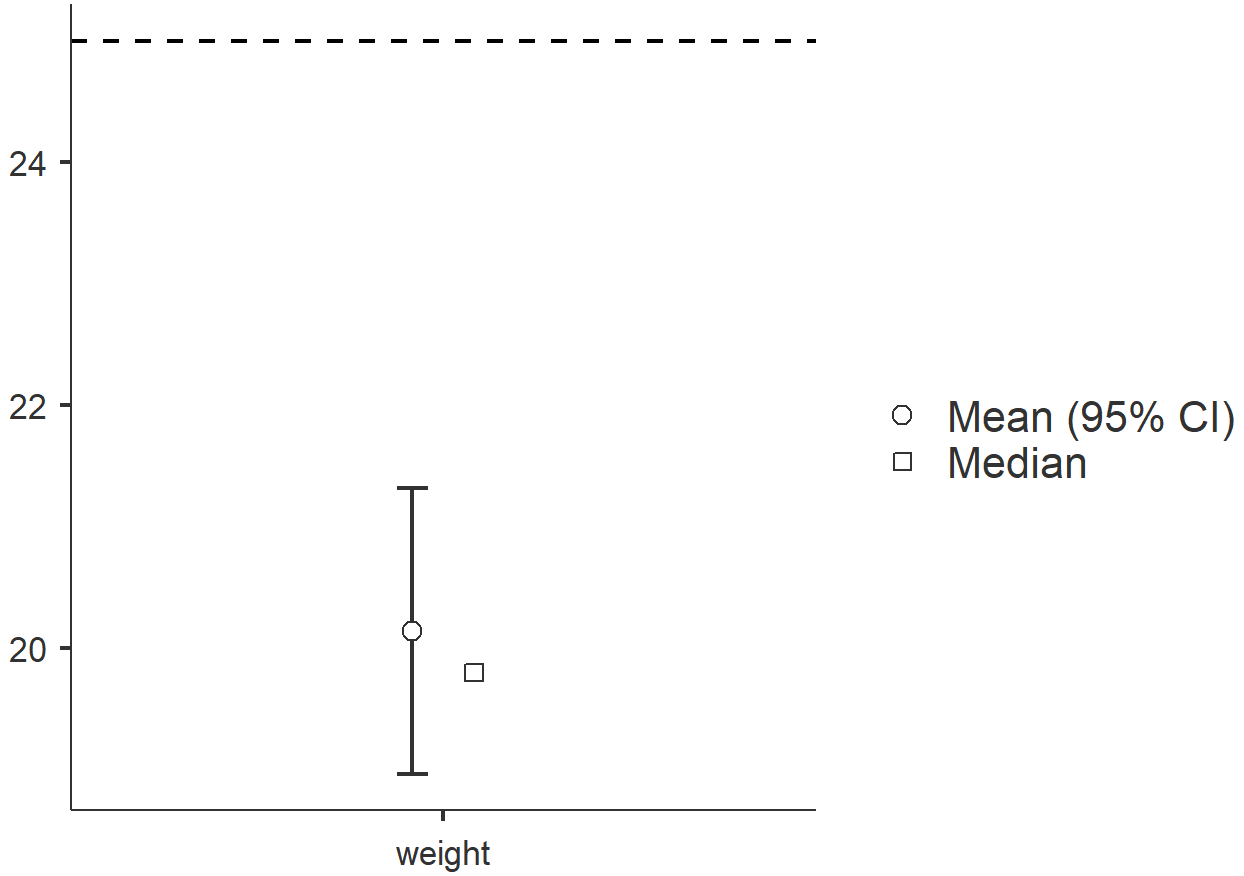
```
##
##  ONE SAMPLE T-TEST
##
##  One Sample T-Test
##  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
##                              Statistic    df          p                        Effect Size
Lower        Upper
##  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
##    weight    Student's t    -8.104504    9.000000    0.0000200    Cohen's d      -2.562869
-3.866962    -1.230830
##  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
##    Note. Ha μ ≠ 25
##
##
##  Normality Test (Shapiro-Wilk)
##  ──────────────────────────────────────
##              W           p
##  ──────────────────────────────────────
##    weight    0.9229514    0.3822300
##  ──────────────────────────────────────
##    Note. A low p-value suggests a
##    violation of the assumption of
##    normality
##
##
##  Descriptives
##  ──────────────────────────────────────────────────────────────────────
##              N     Mean        Median      SD          SE
##  ──────────────────────────────────────────────────────────────────────
##    weight    10    20.14000    19.80000    1.896312    0.5996666
##  ──────────────────────────────────────────────────────────────────────
```

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## ℹ Please use the `linewidth` argument instead.
## ℹ The deprecated feature was likely used in the jmvcore package.
##   Please report the issue at <https://github.com/jamovi/jmvcore/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



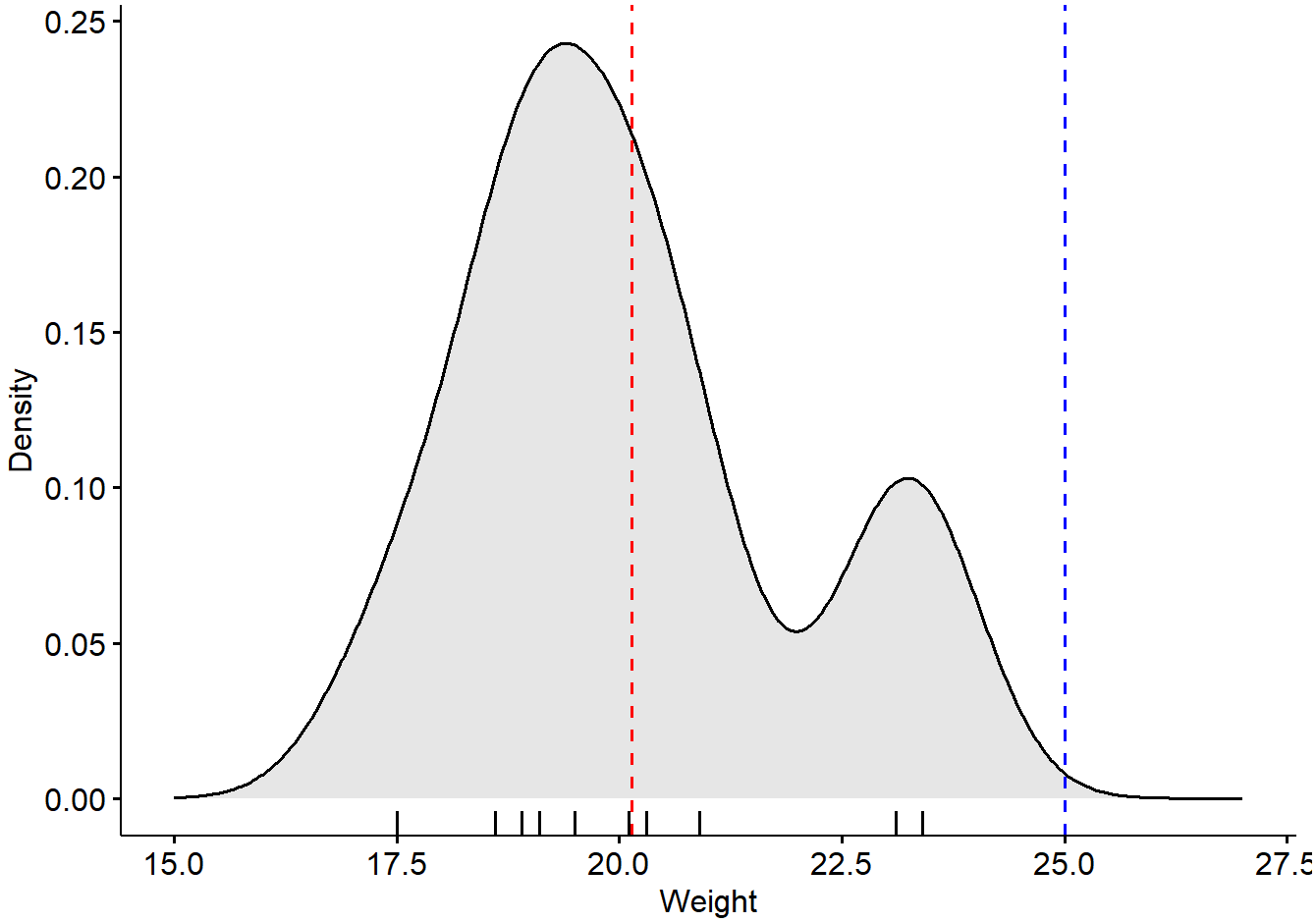

```
mice %>%
ttestOneS(vars=weight,
          testValue = 25,
          hypothesis = 'lt',
          norm = TRUE,
          ci = TRUE,
          effectSize = TRUE,
          ciES = TRUE,
          desc = TRUE,
          plots = TRUE)
```

```
##
##  ONE SAMPLE T-TEST
##
##  One Sample T-Test
##  ──────────────
```



Publishable Plot | Using `rstatix` and `ggpubr` packages

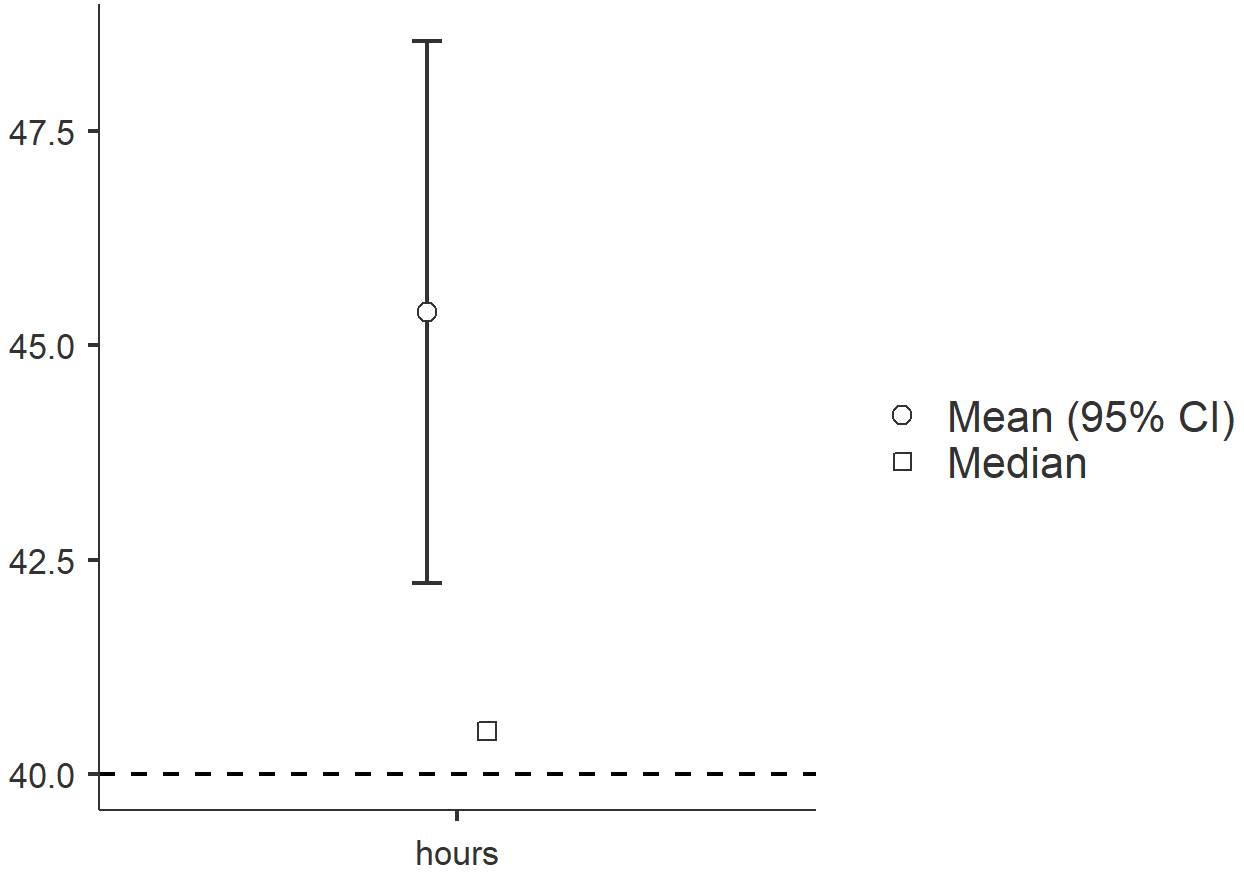
```
# Density Plot

ggdensity(mice,
          x = "weight",
          rug = TRUE,
          fill = "lightgray") +
  scale_x_continuous(limits = c(15, 27)) +
  stat_central_tendency(type = "mean", color = "red", linetype = "dashed") +
  geom_vline(xintercept = 25, color = "blue", linetype = "dashed") +
  labs(x = "Weight",
       y = "Density")
```



Research Question 2

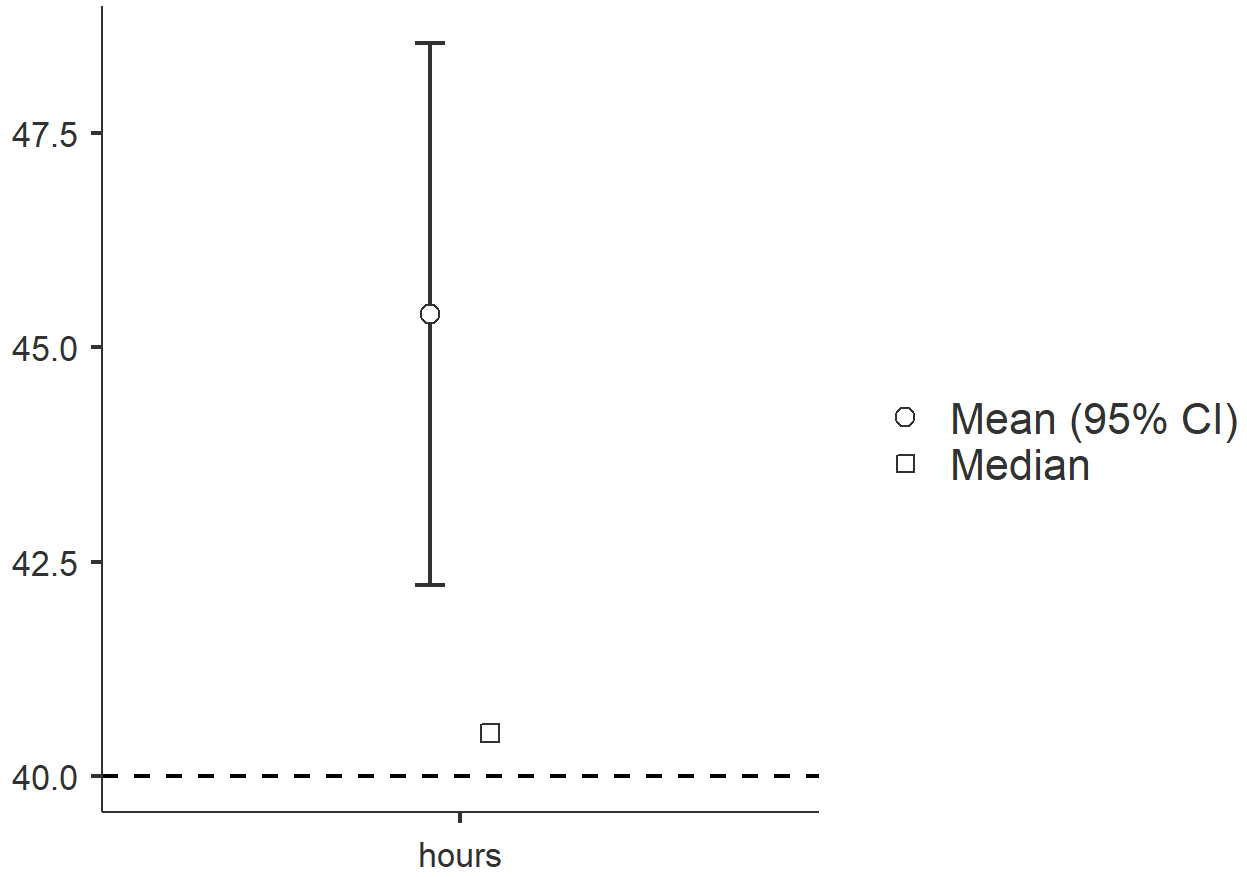
```
gss %>%
ttestOneS(vars=hours,
          testValue = 40,
          hypothesis = 'dt',
          norm = TRUE,
          ci = TRUE,
          effectSize = TRUE,
          ciES = TRUE,
          desc = TRUE,
          plots = TRUE)
```

```
##
##  ONE SAMPLE T-TEST
##
##  One Sample T-Test
##  ──────────────
```



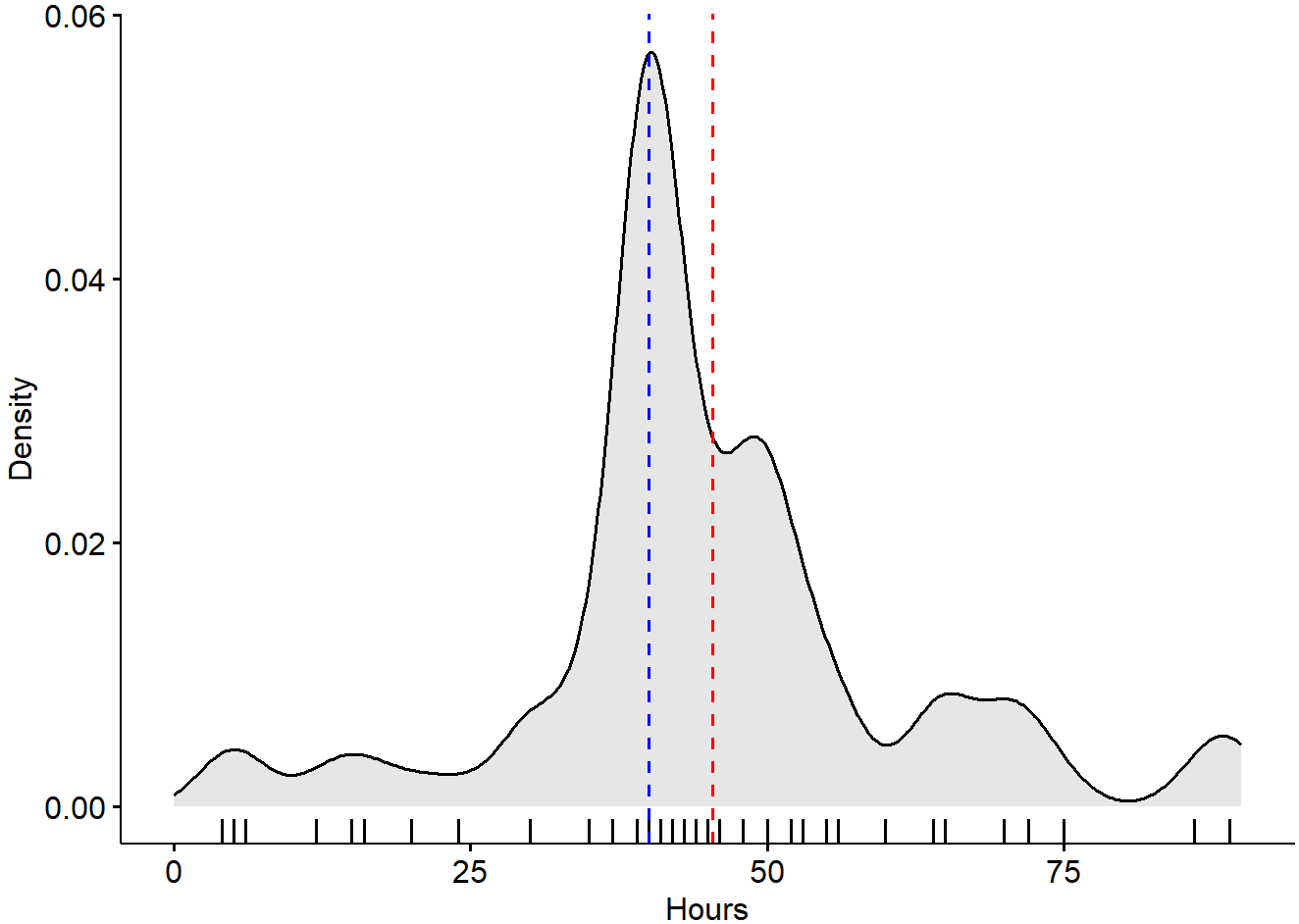

```
gss %>%
ttestOneS(vars=hours,
          testValue = 40,
          hypothesis = 'gt',
          norm = TRUE,
          ci = TRUE,
          effectSize = TRUE,
          ciES = TRUE,
          desc = TRUE,
          plots = TRUE)
```

```
##
##  ONE SAMPLE T-TEST
##
##  One Sample T-Test
##  ──────────────
```




```
# Violations of the Shapiro-Wilks test can set `wilcoxon = TRUE` argument
# gss %>%
# ttestOneS(vars=hours,
#           testValue = 40,
#           hypothesis = 'dt',
#           norm = TRUE,
#           ci = TRUE,
#           effectSize = TRUE,
#           ciES = TRUE,
#           desc = TRUE,
#           wilcoxon = TRUE)
#
# gss %>%
# ttestOneS(vars=hours,
#           testValue = 40,
#           hypothesis = 'gt',
#           norm = TRUE,
#           ci = TRUE,
#           effectSize = TRUE,
#           ciES = TRUE,
#           desc = TRUE,
#           wilcoxon = TRUE)
```

```
# Density Plot

ggdensity(gss,
          x = "hours",
          rug = TRUE,
          fill = "lightgray") +
  scale_x_continuous(limits = c(0, 90)) +
  stat_central_tendency(type = "mean", color = "red", linetype = "dashed") +
  geom_vline(xintercept = 40, color = "blue", linetype = "dashed") +
  labs(x = "Hours",
       y = "Density")
```



APA Style Results Section

Statistical Interpretation:

A one-sample *t*-test was performed to evaluate whether there was a difference between the weight in a sample of mice (*n* = 10) compared to the general population of mice ($\mu = 25$). The mean weight in the sample of mice (M = 20.14, SD = 1.90) was statistically significantly different than the average weight of mice in the general population, *t*(9) = -8.11, *p* < .001, Cohen's *d* = -2.56, 95% CI [-3.87, -1.23].

Paired-Sample (Repeated-Measure) `t`-test

Numerical variable collected twice OR `within-subject` DV

Research Question 1: Does the weight of 10 mice differ from `Before` to `After` following a prescribed treatment? Does the treatment lead to increased weight from `Before` to `After` for the 10 mice?

```
# Load and inspect demo data set(s)

# Load `mice2` data from `datarium` package
data(mice2, package="datarium")
head(mice2)
```

```
##   id before after
## 1  1  187.2 429.5
## 2  2  194.2 404.4
## 3  3  231.7 405.6
## 4  4  200.5 397.2
## 5  5  201.7 377.9
## 6  6  235.0 445.8
```

```
# Transform into long data:
# gather the before and after values in the same column
mice2_long <- mice2 %>%
  gather(key = "group", value = "weight", before, after)
head(mice2_long)
```

```
##   id  group weight
## 1  1 before  187.2
## 2  2 before  194.2
## 3  3 before  231.7
## 4  4 before  200.5
## 5  5 before  201.7
## 6  6 before  235.0
```

Hypothesis Statement: 2-tailed (Non-directional)

H_0: Null Hypothesis — Average weight of sampled mice is not different due to treatment (i.e., from `Before` to `After`), $\mu_{Before} = \mu_{After}$ OR $\mu_{Before-After} = 0$

H_a: Alternate Hypothesis — Average weight of sampled mice is different due to treatment (i.e., from `Before` to `After`), $\mu_{Before} \neq \mu_{After}$ OR $\mu_{Before-After} \neq 0$

Hypothesis Statement: 1-tailed (Directional)

H_0: Null Hypothesis — Average weight of sampled mice is not greater following treatment (i.e., from `Before` to `After`), $\mu_{Before} \geq \mu_{After}$ OR $\mu_{Before-After} \geq 0$

H_a: Alternate Hypothesis — Average weight of sampled mice is greater following treatment (i.e., from `Before` to `After`), $\mu_{Before} < \mu_{After}$ OR $\mu_{Before-After} < 0$

Research Question 2:

```
# Load and inspect demo data set(s)

# Load `depression` data from `datarium` package
data(depression, package = "datarium")
depression<- depression %>%
  select(id,t0,t3)
head(depression)
```

```
## # A tibble: 6 × 3
##   id       t0    t3
##   <fct> <dbl> <dbl>
## 1 1       296   242
## 2 2       376   126
## 3 3       309   173
## 4 4       222   135
## 5 5       150   266
## 6 6       316   194
```

```
# Transform into long data:
# gather the before and after values in the same column
dep_long <- depression %>%
  gather(key = "time_period", value = "depression", t0, t3)
head(dep_long)
```

```
## # A tibble: 6 × 3
##   id    time_period depression
##   <fct> <chr>            <dbl>
## 1 1     t0                 296
## 2 2     t0                 376
## 3 3     t0                 309
## 4 4     t0                 222
## 5 5     t0                 150
## 6 6     t0                 316
```

Hypothesis Statement: 2-tailed (Non-directional)

H_0: Null Hypothesis — Average depression level of sampled patients is not different from `Baseline` to `Follow-Up`, $\mu_{Baseline} = \mu_{Follow-Up}$ OR $\mu_{Baseline-Follow-Up} = 0$

H_a: Alternate Hypothesis — Average depression level of sampled patients is different from `Baseline` to `Follow-Up`, $\mu_{Baseline} \neq \mu_{Follow-Up}$ OR $\mu_{Baseline-Follow-Up} \neq 0$

Hypothesis Statement: 1-tailed (Directional)

H_0: Null Hypothesis — Average depression level of sampled patients is not lower from `Baseline` to `Follow-Up`, $\mu_{Baseline} \geq \mu_{Follow-Up}$ OR $\mu_{Baseline-Follow-Up} \geq 0$

H_a: Alternate Hypothesis — Average depression level of sampled patients is lower from `Baseline` to `Follow-Up`, $\mu_{Baseline} < \mu_{Follow-Up}$ OR $\mu_{Baseline-Follow-Up} < 0$

Descriptive Statistics (Do not forget to visualize!!)

```
# Change `group` into a factor
mice2_long$group<-factor(mice2_long$group)

# Use the `describeBy` function to isolate `weight`
describeBy(mice2_long[,3],group=mice2_long$group,mat=TRUE)
```

```
##     item group1 vars  n   mean       sd median trimmed      mad   min   max
## X11    1  after    1 10 400.04 30.08688 405.00 402.200 25.05594 337.0 445.8
## X12    2 before    1 10 200.56 20.02816 197.35 199.775 15.93795 172.4 235.0
##     range       skew  kurtosis       se
## X11 108.8 -0.5285935 -0.391983 9.514307
## X12  62.6  0.5132390 -1.077594 6.333460
```

```
# Change `time_period` into a factor
dep_long$time_period<-factor(dep_long$time_period)

# Use the `describeBy` function to isolate `depression`
describeBy(dep_long[,3],group=dep_long$time_period,mat=TRUE)
```

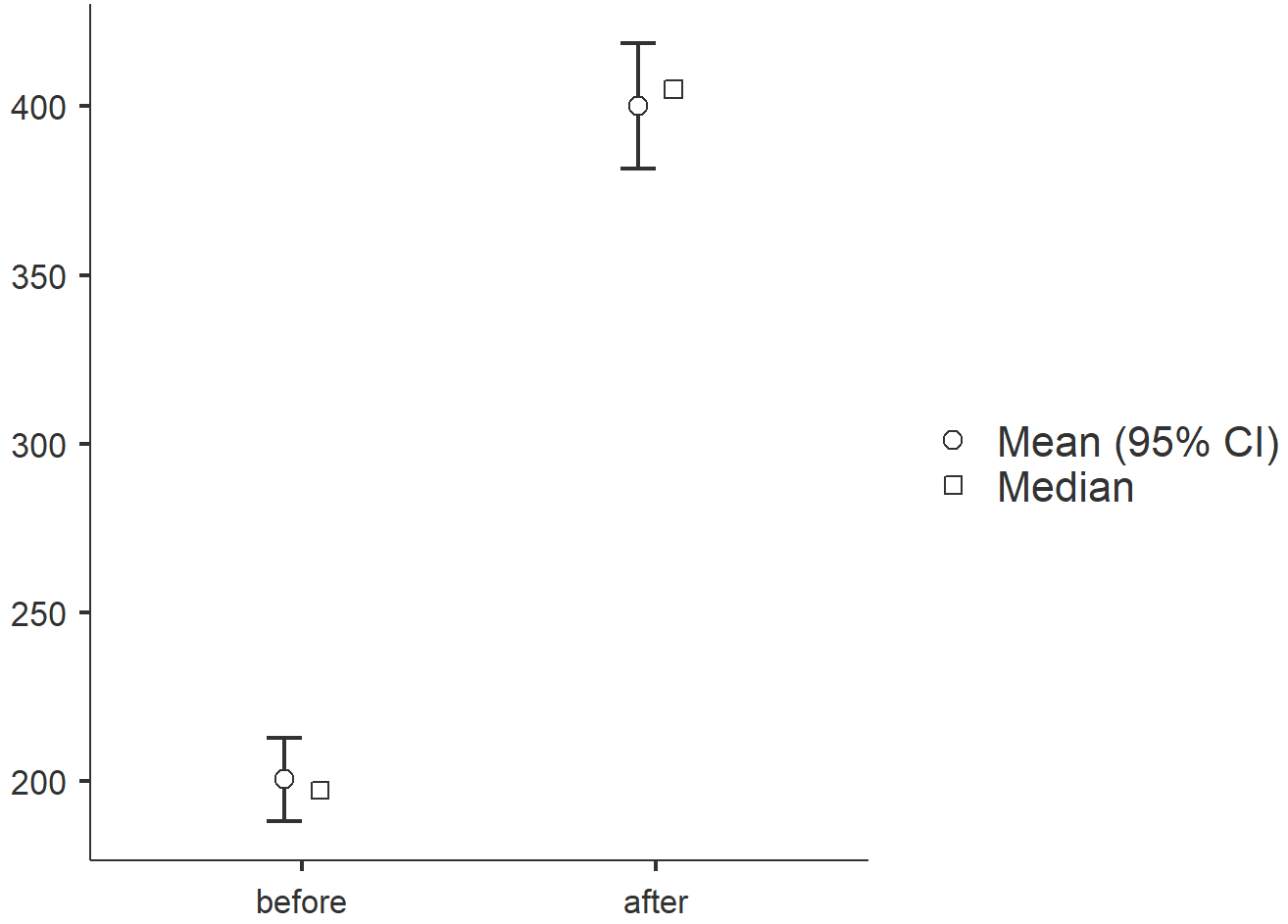
```
##             item group1 vars  n     mean       sd median trimmed     mad min
## depression1    1     t0    1 24 292.3750 73.86757  309.5   295.3 41.5128 138
## depression2    2     t3    1 24 151.9167 78.02002  137.0   150.2 54.1149   6
##             max range       skew    kurtosis       se
## depression1 447   309 -0.4958564 -0.00301943 15.07816
## depression2 358   352  0.5007439  0.50394014 15.92577
```

`jmv` Paired-Sample `t`-test `ttestPS()`

Research Question 1

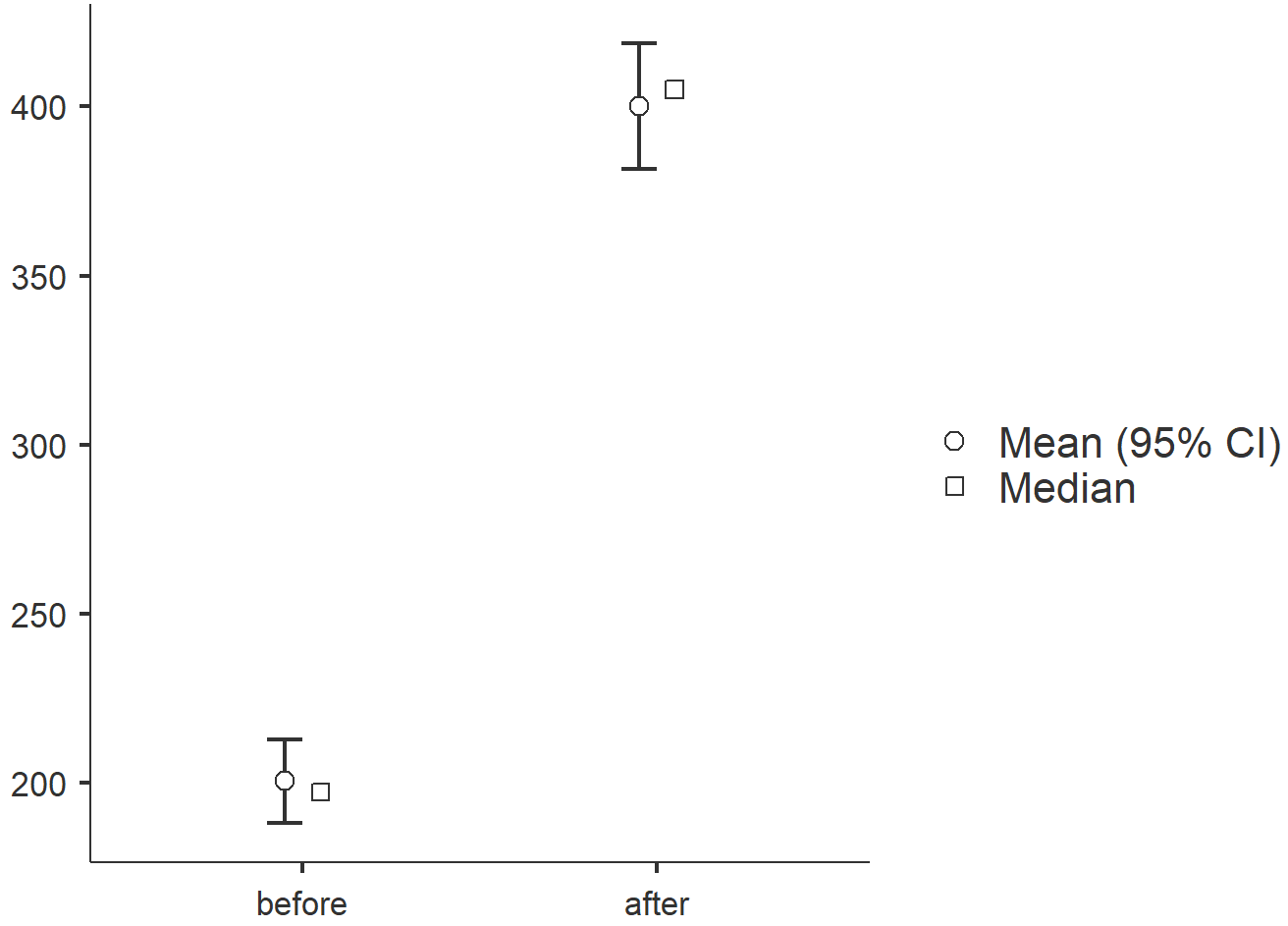
```
mice2 %>%
ttestPS(pairs=list(list(i1 = "before", i2 = "after")),
        hypothesis = 'different',
        norm = TRUE,
        ci = TRUE,
        effectSize = TRUE,
        ciES = TRUE,
        desc = TRUE,
        plots = TRUE)
```

```
##
##  PAIRED SAMPLES T-TEST
##
##  Paired Samples T-Test
##  ──────────────
```




```
mice2 %>%
ttestPS(pairs=list(list(i1 = "before", i2 = "after")),
        hypothesis = 'twoGreater',
        norm = TRUE,
        ci = TRUE,
        effectSize = TRUE,
        ciES = TRUE,
        desc = TRUE,
        plots = TRUE)
```

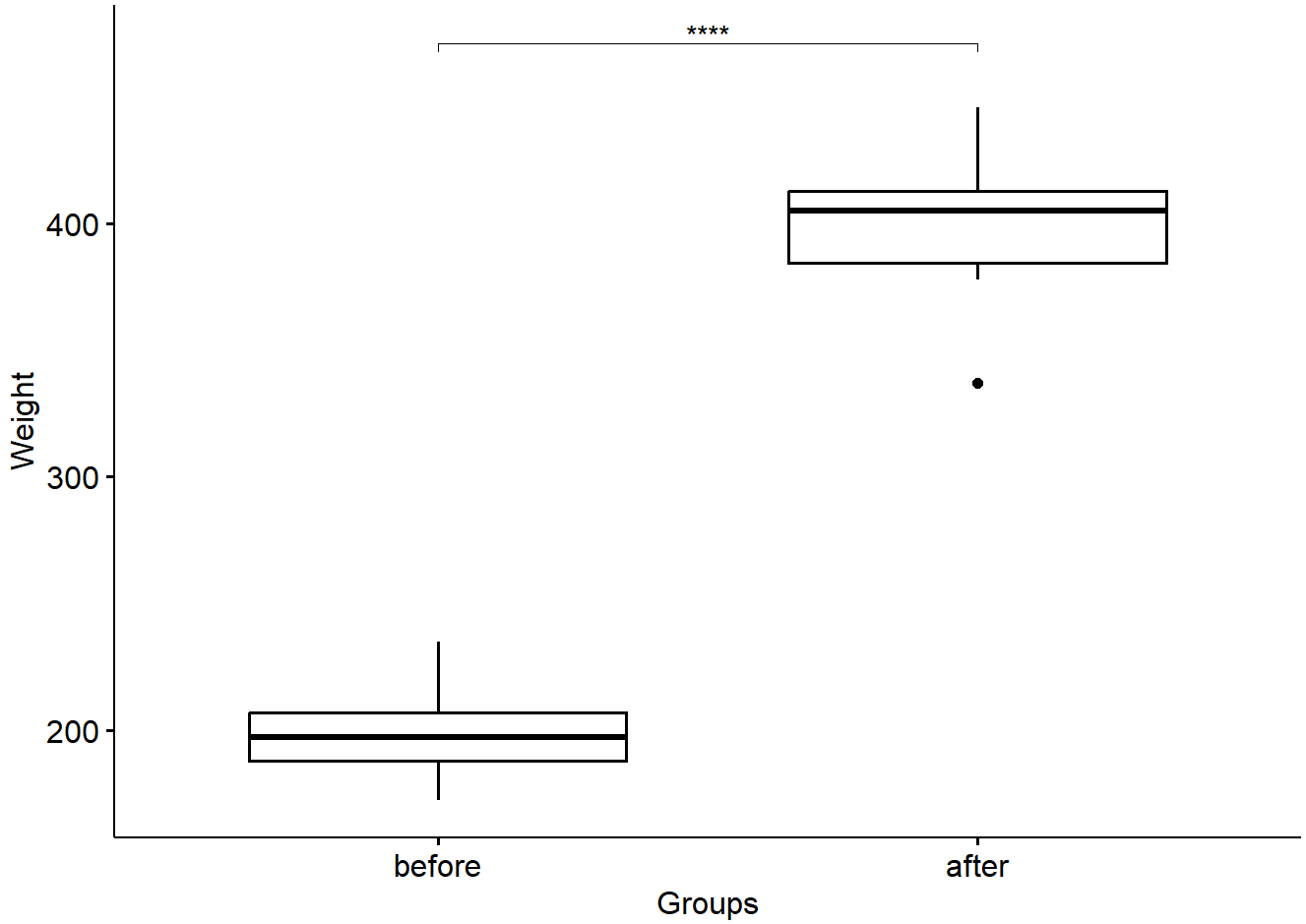
```
##
##  PAIRED SAMPLES T-TEST
##
##  Paired Samples T-Test
##  ──────────────
```



Publishable Plot | Using `rstatix` and `ggpubr`

packages

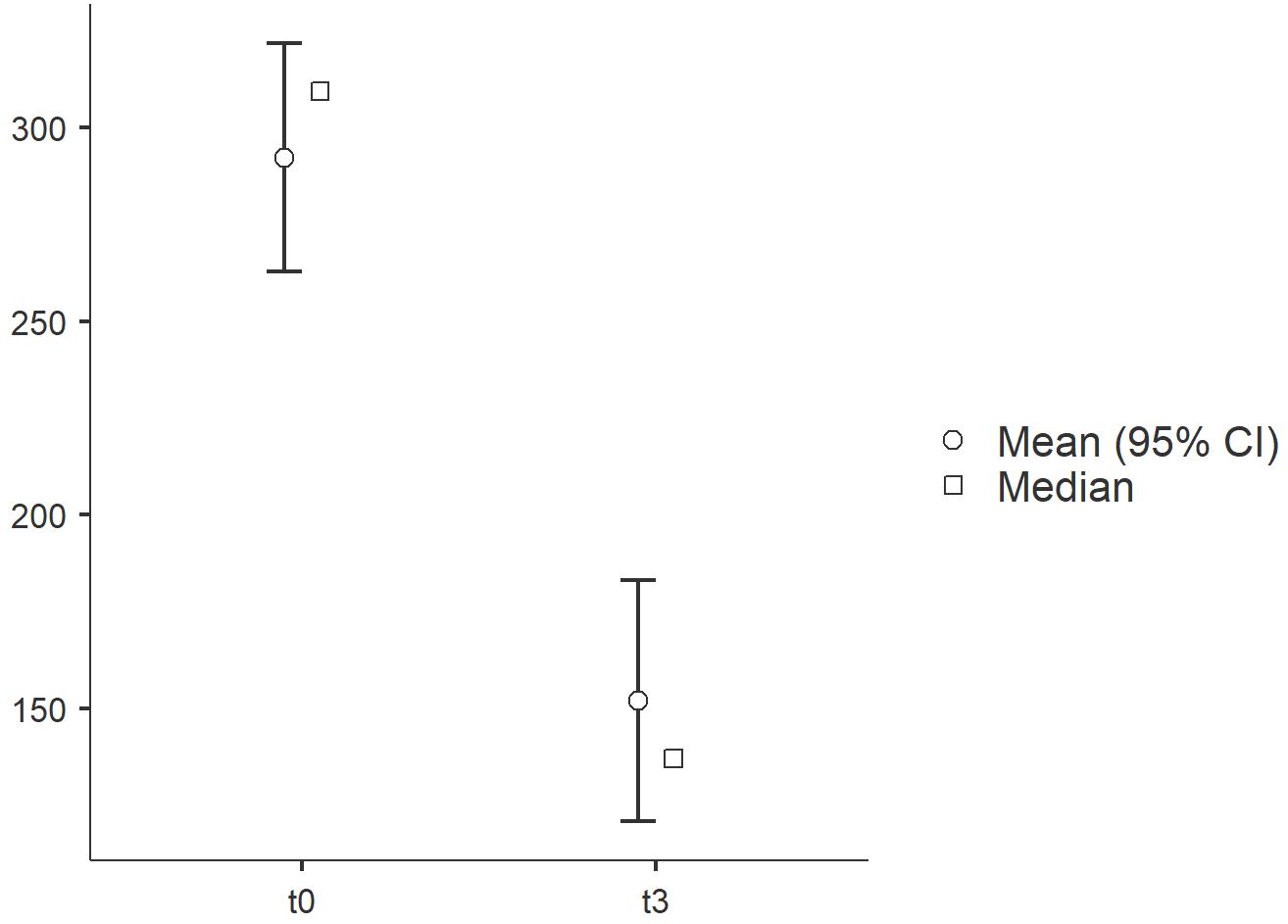
```
# Side-by-Side Boxplot Comparison

# Create a box plot
bxp <- ggboxplot(mice2_long,
                x = "group",
                y = "weight",
                order = c("before", "after"),
                ylab = "Weight", xlab = "Groups")

# Add p-value and significance levels

stat_test <- mice2_long  %>%
  t_test(weight ~ group,
         ref.group = "before",
         paired = TRUE,
         alternative = "two.sided") %>% # `alternative =` can be set to `greater` or `less` as w
ell
  add_significance() %>%
  add_xy_position(x = "group")

# Print boxplots with significance bracket

bxp +
  stat_pvalue_manual(stat_test, tip.length = .01)
```



Research Quesiton 2

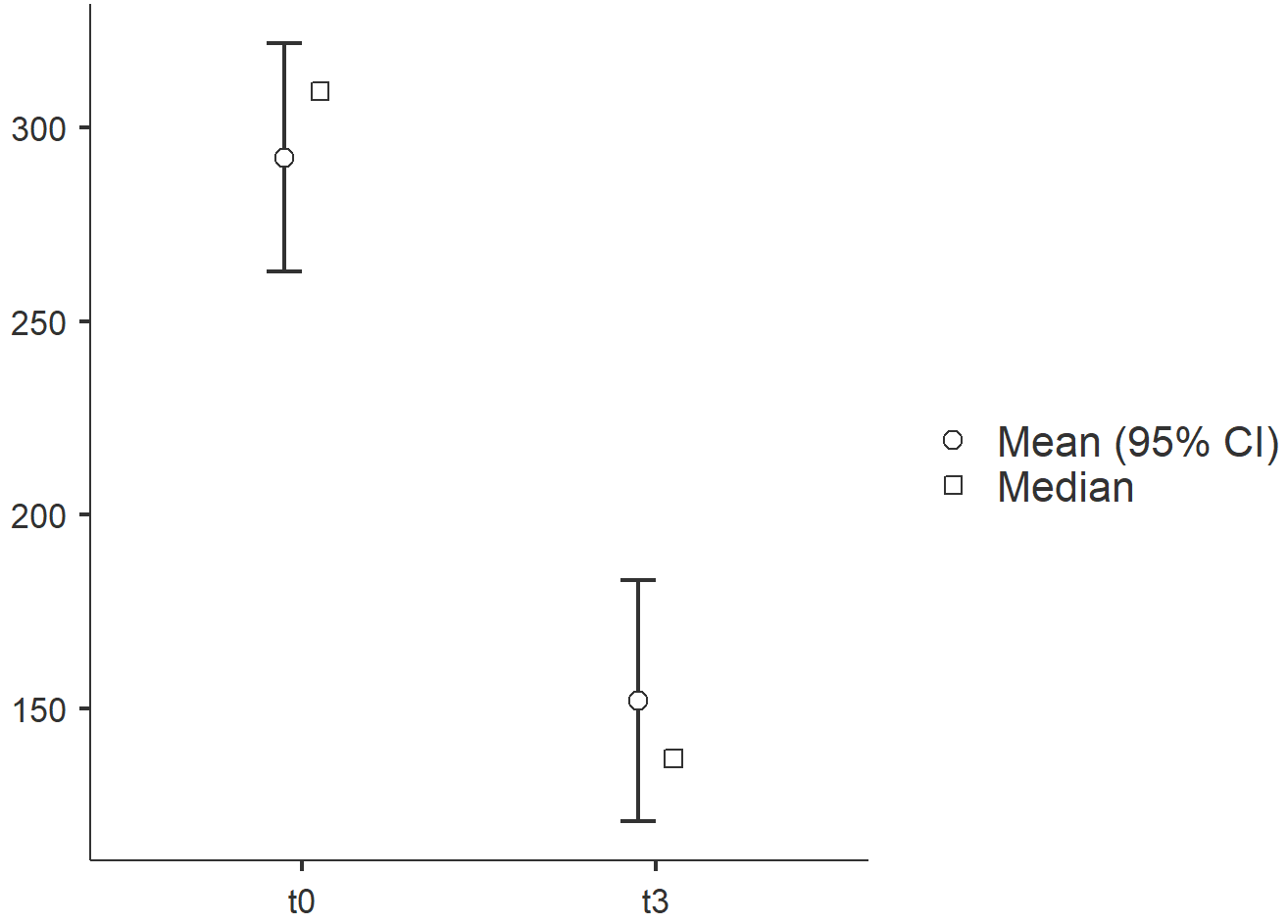
```
depression %>%
ttestPS(pairs=list(list(i1 = "t0", i2 = "t3")),
        hypothesis = 'different',
        norm = TRUE,
        ci = TRUE,
        effectSize = TRUE,
        ciES = TRUE,
        desc = TRUE,
        plots = TRUE)
```

```
##
##  PAIRED SAMPLES T-TEST
##
##  Paired Samples T-Test
##  ──────────────
```

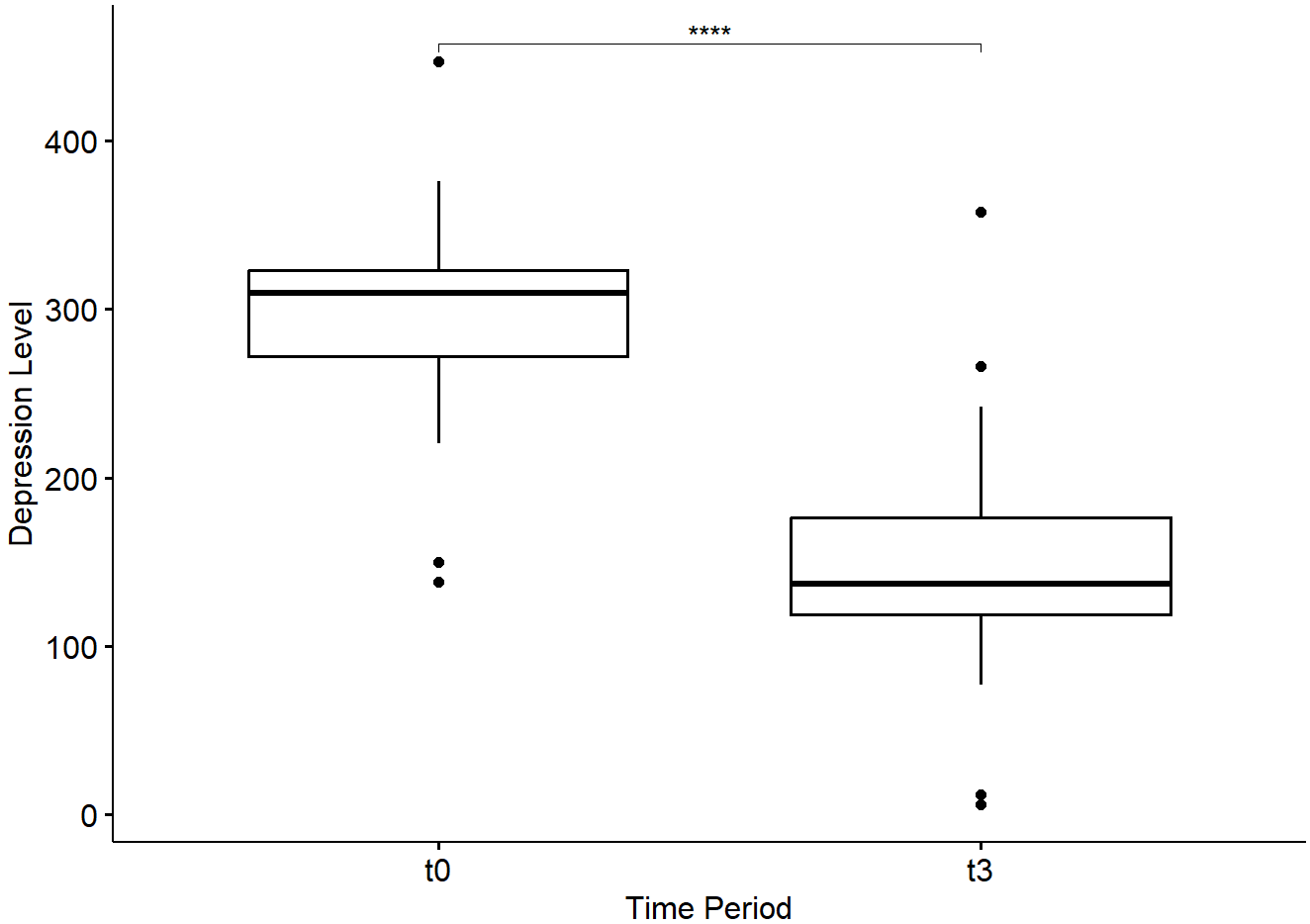



```
depression %>%
ttestPS(pairs=list(list(i1 = "t0", i2 = "t3")),
        hypothesis = 'oneGreater',
        norm = TRUE,
        ci = TRUE,
        effectSize = TRUE,
        ciES = TRUE,
        desc = TRUE,
        plots = TRUE)
```

```
##
##  PAIRED SAMPLES T-TEST
##
##  Paired Samples T-Test
##  ──────────────
```




```
# Side-by-Side Boxplot Comparison

# Create a box plot
bxp <- ggboxplot(dep_long,
                 x = "time_period",
                 y = "depression",
                 order = c("t0", "t3"),
                 ylab = "Depression Level", xlab = "Time Period")

# Add p-value and significance levels
stat_test <- dep_long  %>%
  t_test(depression ~ time_period,
         paired = TRUE,
         ref.group = "t0",
         alternative = "greater") %>%
  add_significance() %>%
  add_xy_position(x = "time_period")

# Print boxplots with significance bracket
bxp +
  stat_pvalue_manual(stat_test, tip.length = 0.01)
```



APA Style Results Section

Statistical Interpretation:

A paired-sample *t*-test was conducted to determine whether there was a difference between the depression level among our sample from baseline (`t0`) to a 6 month follow-up (`t3`). The results indicated that depression level at baseline (*M* = 292.38, *SD* = 73.87) was statistically significantly higher than depression level at 6 month follow-up (*M* = 151.92, *SD* = 78.02), *t*(23) = 6.69, *p* < .001, Cohen's *d* = 1.37, 95% CI [.798, 1.918].

Independent-Samples (Between-Subject) `t`-test

Numerical DV; Mutually Exclusive, Binary Grouping Variable

Research Question 1: Does the average stress level of 60 participants differ due to a randomly assigned stress treatment ($n_{perGroup} = 30$)? Was the average stress level of the `treatment group` lower than the average stress level of the `non-treatment group` ?

```
# Load and inspect demo data set(s)

# Load `mice2` data from `datarium` package
data(stress, package="datarium")
stress <- stress %>%
  select(id,score,treatment)
head(stress)
```

```
## # A tibble: 6 × 3
##      id score treatment
##   <int> <dbl> <fct>
## 1     1  95.6 yes
## 2     2  82.2 yes
## 3     3  97.2 yes
## 4     4  96.4 yes
## 5     5  81.4 yes
## 6     6  83.6 yes
```

Hypothesis Statement: 2-tailed (Non-directional)

H_0: Null Hypothesis — Average stress level was not different between treatment groups (i.e., from `Treatment` to `Non-treatment`), $\mu_{Treatment} = \mu_{Non-Treatment}$ OR $\mu_{Treatment} - \mu_{Non-Treatment} = 0$

H_a: Alternate Hypothesis — Average stress level was different between treatment groups (i.e., from `Treatment` to `Non-treatment`), $\mu_{Treatment} \neq \mu_{Non-Treatment}$ OR $\mu_{Treatment} - \mu_{Non-Treatment} \neq 0$

Hypothesis Statement: 1-tailed (Directional)

H_0: Null Hypothesis — Average stress level was not lower between the `treatment` and `non-treatment` groups, $\mu_{Treatment} \geq \mu_{Non-Treatment}$ OR $\mu_{Treatment} - \mu_{Non-Treatment} \geq 0$

H_a: Alternate Hypothesis — Average stress level was lower between the `treatment` and `non-treatment` groups, $\mu_{Treatment} < \mu_{Non-Treatment}$ OR $\mu_{Treatment} - \mu_{Non-Treatment} < 0$

Research Question 2: Does the average increase in hours of sleep across 20 participants differ due to a randomly assigned treatment ($n_{perGroup} = 10$)? Was the average increase in hours of sleep of the `medication group` greater than the average increase in hours of sleep of the `Control group`?

```
# Load and inspect demo data set(s)

# Load `depression` data from `datarium` package
data(sleep, package = "datasets")
head(sleep)
```

```
##   extra group ID
## 1   0.7     1  1
## 2  -1.6     1  2
## 3  -0.2     1  3
## 4  -1.2     1  4
## 5  -0.1     1  5
## 6   3.4     1  6
```

Hypothesis Statement: 2-tailed (Non-directional)

H_0: Null Hypothesis — Average increases in hours of sleep was not different between treatment groups (i.e., from `Medication` to `Control`), $\mu_{Medication} = \mu_{Control}$ OR $\mu_{Medication} - \mu_{Control} = 0$

H_a: Alternate Hypothesis — Average increases in hours of sleep was different between treatment groups (i.e., from `Medication` to `Control`), $\mu_{Medication} \neq \mu_{Control}$ OR $\mu_{Medication} - \mu_{Control} \neq 0$

Hypothesis Statement: 1-tailed (Directional)

H_0: Null Hypothesis — Average increases in hours of sleep was not greater from `Medication` to `Control Treatment Groups` , $\mu_{Medication} \leq \mu_{Control}$ OR $\mu_{Medication} - \mu_{Control} \leq 0$

H_a: Alternate Hypothesis — Average increases in hours of sleep was greater from `Medication` to `Control Treatment Groups` , $\mu_{Medication} > \mu_{Control}$ OR $\mu_{Medication} - \mu_{Control} > 0$

Descriptive Statistics (Do not forget to visualize!!)

```
# Use the `describeBy` function
describeBy(stress$score,group=stress$treatment,mat=TRUE)
```

```
##     item group1 vars  n     mean       sd median  trimmed     mad  min   max
## X11    1    yes    1 30 82.15667 9.161527  83.45 82.08750 9.11799 65.8  97.2
## X12    2     no    1 30 86.99667 6.166678  87.30 87.18333 7.11648 75.3 100.0
##     range        skew   kurtosis       se
## X11  31.4 -0.05540563 -1.1109814 1.672658
## X12  24.7 -0.14771494 -0.6624365 1.125876
```

```
describeBy(sleep$extra,group=sleep$group,mat=TRUE)
```

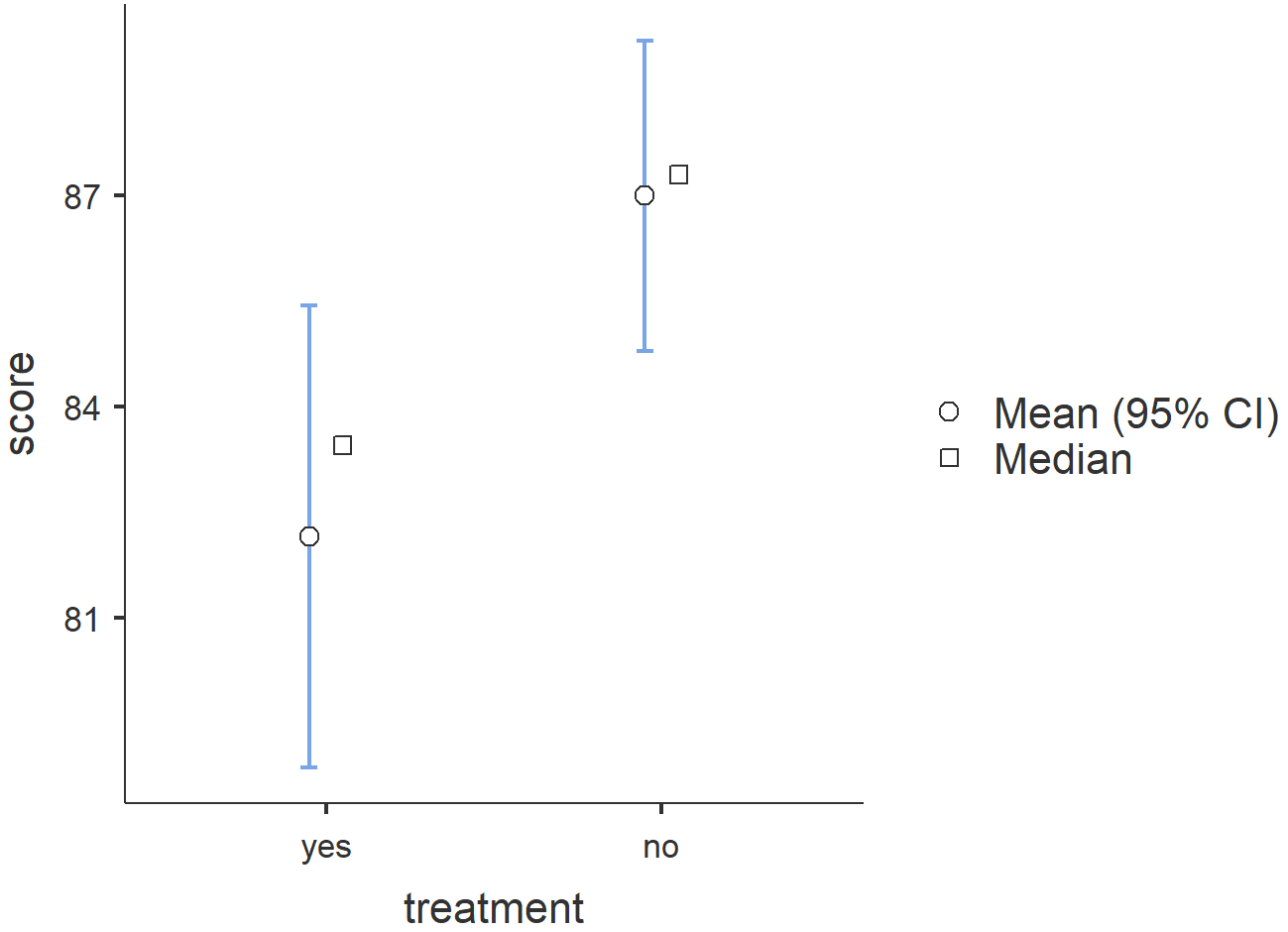
```
##     item group1 vars  n mean       sd median trimmed     mad  min max range
## X11    1      1    1 10 0.75 1.789010   0.35  0.6750 1.55673 -1.6 3.7   5.3
## X12    2      2    1 10 2.33 2.002249   1.75  2.2375 2.44629 -0.1 5.5   5.6
##          skew  kurtosis        se
## X11 0.4182629 -1.300391 0.5657345
## X12 0.2777800 -1.662971 0.6331666
```

`jmv` Independent-Samples `t`-test `ttestIS()`

Research Question 1

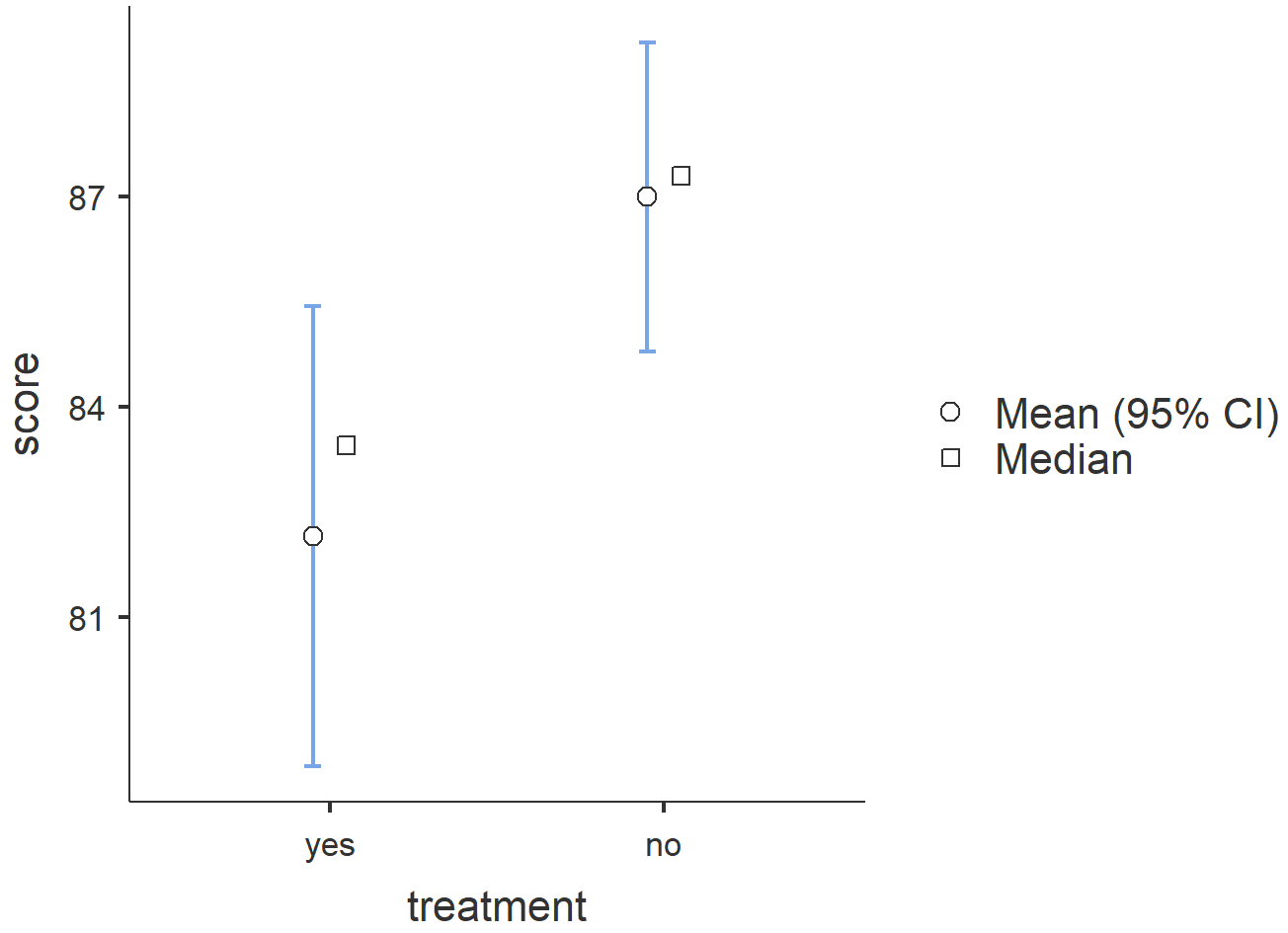
```
stress %>%
ttestIS(vars= "score",
        group = "treatment",
        hypothesis = 'different',
        norm = TRUE,
        eqv = TRUE,
        ci = TRUE,
        effectSize = TRUE,
        ciES = TRUE,
        desc = TRUE,
        plots = TRUE)
```

```
##
##  INDEPENDENT SAMPLES T-TEST
##
##  Independent Samples T-Test
##  ──────────────
```




```
stress %>%
ttestIS(vars= "score",
        group = "treatment",
        hypothesis = 'twoGreater',
        norm = TRUE,
        eqv = TRUE,
        ci = TRUE,
        effectSize = TRUE,
        ciES = TRUE,
        desc = TRUE,
        plots = TRUE)
```

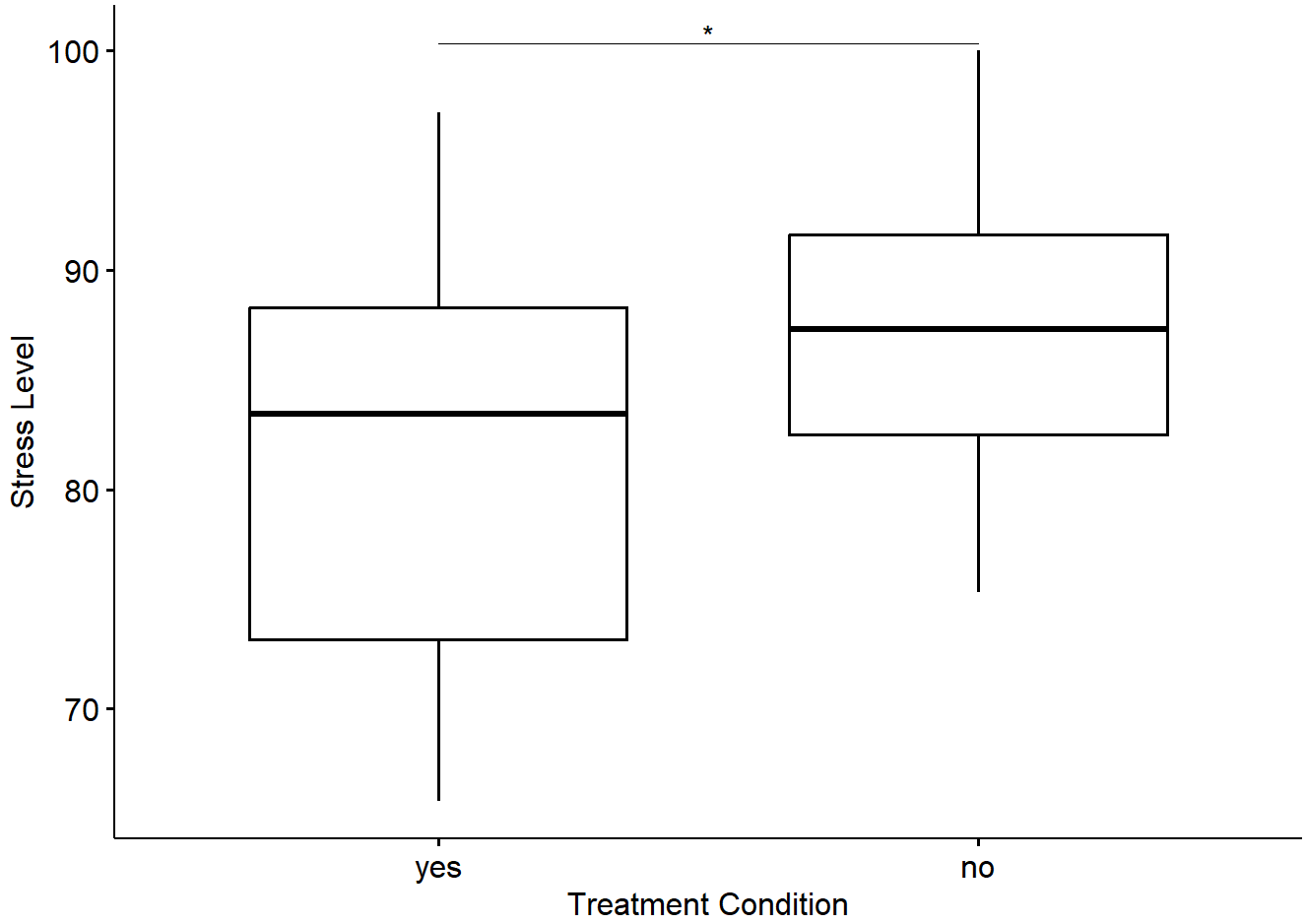
```
##
##  INDEPENDENT SAMPLES T-TEST
##
##  Independent Samples T-Test
##  ──────────────
```




```
# Violations of the Levenes test can set `welchs = TRUE` argument
# stress %>%
# ttestIS(vars= "score",
#         group = "treatment",
#         hypothesis = 'different',
#         norm = TRUE,
#         eqv = TRUE,
#         ci = TRUE,
#         effectSize = TRUE,
#         ciES = TRUE,
#         desc = TRUE,
#         welchs = TRUE)
#
# stress %>%
# ttestIS(vars= "score",
#         group = "treatment",
#         hypothesis = 'twoGreater',
#         norm = TRUE,
#         eqv = TRUE,
#         ci = TRUE,
#         effectSize = TRUE,
#         ciES = TRUE,
#         desc = TRUE,
#         welchs = TRUE)
```

Publishable Plot | Using `rstatix` and `ggpubr` packages

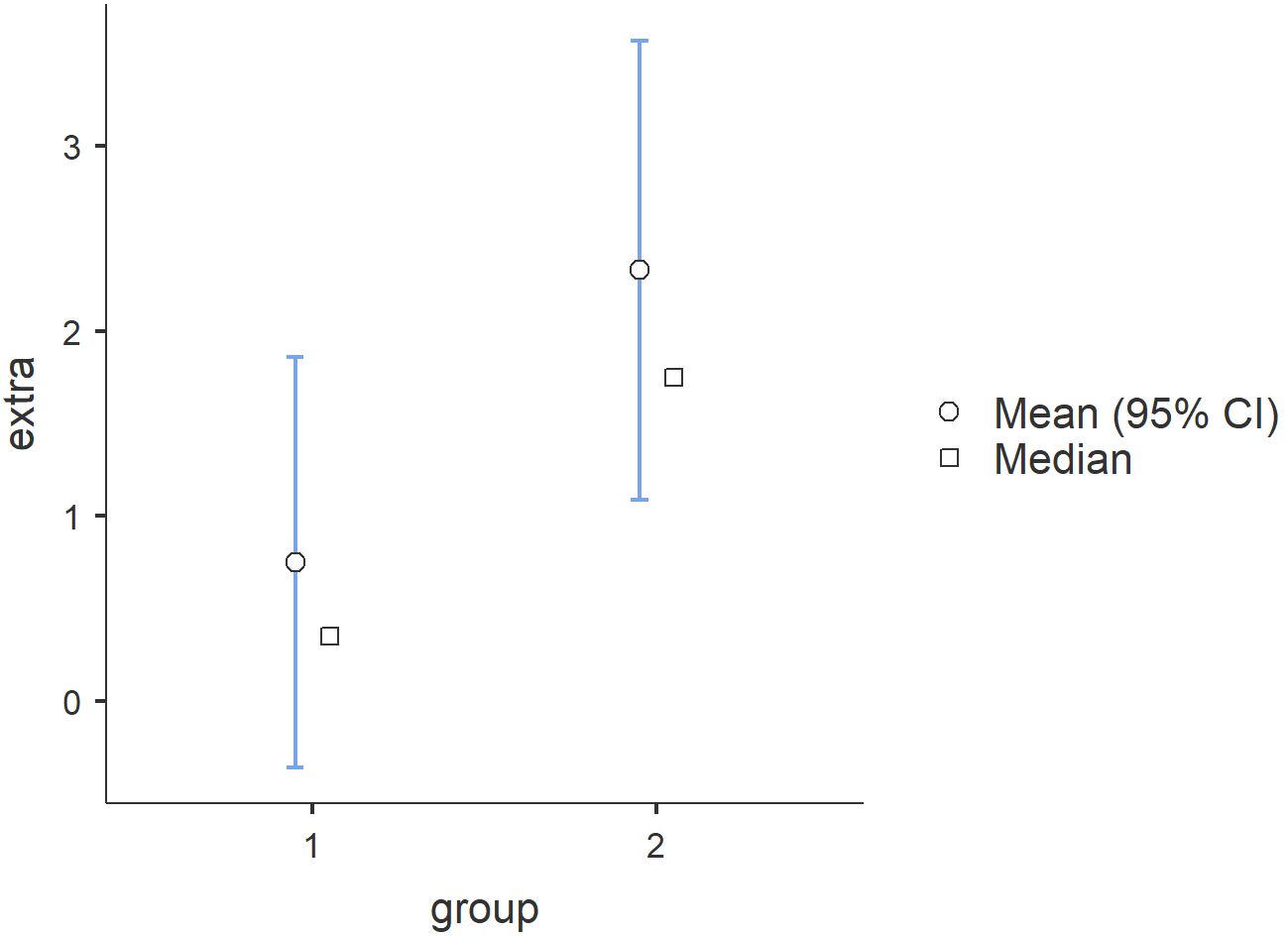
```
# Side-by-Side Boxplot Comparison

# Create a box plot
bxp <- ggboxplot(stress,
                x = "treatment",
                y = "score",
                order = c("yes", "no"),
                ylab = "Stress Level", xlab = "Treatment Condition")

# Add p-value and significance levels
stat_test <- stress  %>%
  t_test(score ~ treatment,
         ref.group = "yes",
         alternative = "two.sided",
         var.equal = TRUE) %>%
  add_significance() %>%
  add_xy_position(x = "treatment")


# Print boxplots with significance bracket
bxp +
  stat_pvalue_manual(stat_test, tip.length = 0)
```



Research Question 2

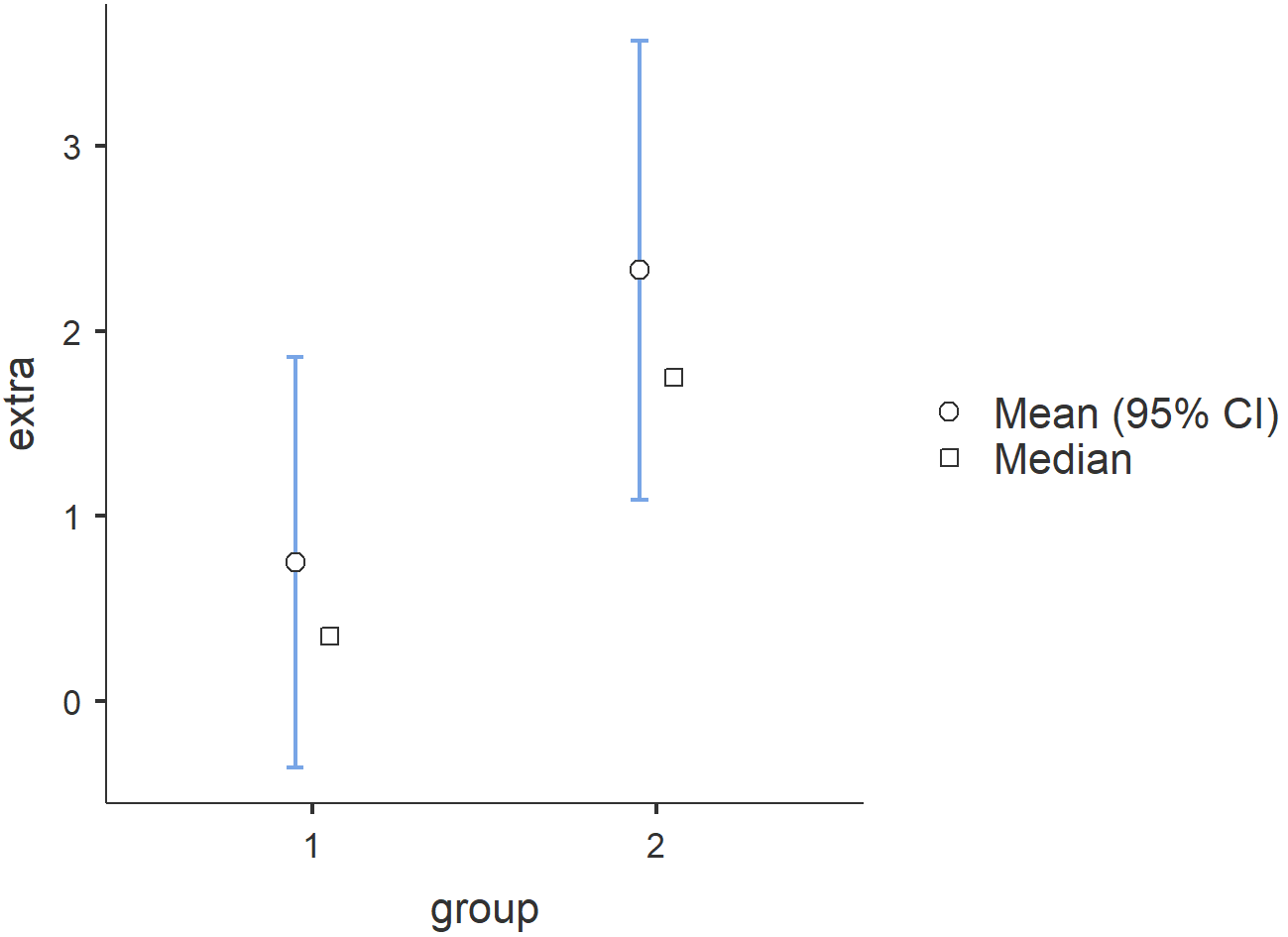
```
sleep %>%
ttestIS(vars= "extra",
        group = "group",
        hypothesis = 'different',
        norm = TRUE,
        eqv = TRUE,
        ci = TRUE,
        effectSize = TRUE,
        ciES = TRUE,
        desc = TRUE,
        plots = TRUE)
```

```
##
##  INDEPENDENT SAMPLES T-TEST
##
##  Independent Samples T-Test
##  ─────────────
```

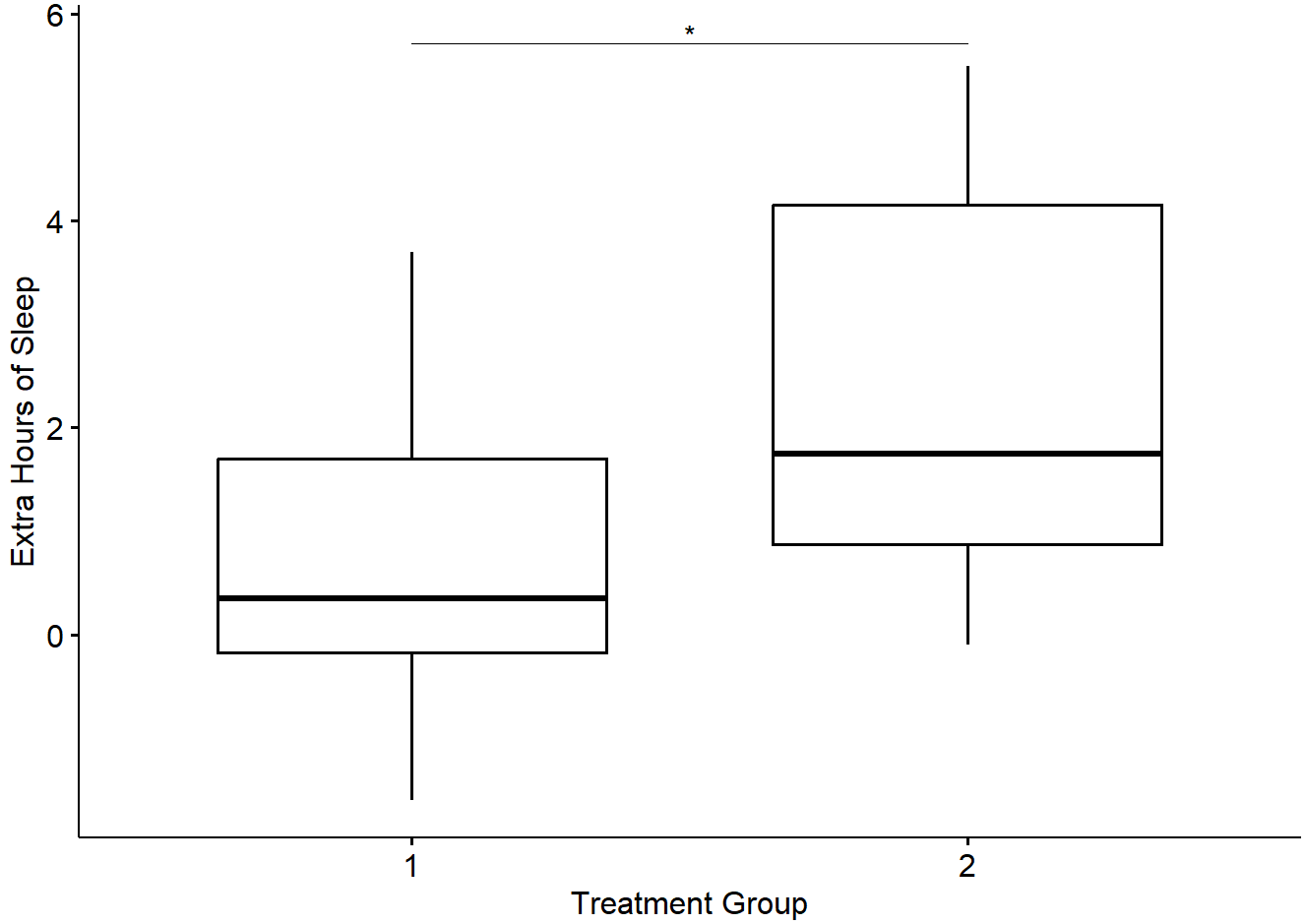



```
sleep %>%
ttestIS(vars= "extra",
        group = "group",
        hypothesis = 'twoGreater',
        norm = TRUE,
        eqv = TRUE,
        ci = TRUE,
        effectSize = TRUE,
        ciES = TRUE,
        desc = TRUE,
        plots = TRUE)
```

```
##
##  INDEPENDENT SAMPLES T-TEST
##
##  Independent Samples T-Test
##  ─────────────
```




```
# Side-by-Side Boxplot Comparison

# Create a box plot
bxp <- ggboxplot(sleep,
                x = "group",
                y = "extra",
                order = c("1", "2"),
                ylab = "Extra Hours of Sleep", xlab = "Treatment Group")

# Add p-value and significance levels
stat_test <- sleep  %>%
  t_test(extra ~ group,
         ref.group = "1",
         alternative = "less",
         var.equal = TRUE) %>%
  add_significance() %>%
  add_xy_position(x = "group")

# Print boxplots with significance bracket
bxp +
  stat_pvalue_manual(stat_test, tip.length = 0)
```



APA Style Results Section

Statistical Interpretation:

An independent samples *t*-test was conducted to determine whether there was a difference between the extra hours of sleep obtained by participants who received the experimental medication and participants who received the placebo. The results indicated that participants who received the medication (M = 2.33, SD = 2.00) did not have statistically significantly greater levels of extra hours of sleep compared to participants who received the placebo (M = 0.75, SD = 1.79), $t(18) = -1.86$, $p = .079$, Cohen's $d = -0.83$, 95% CI [-1.739, 0.096].